Yu (Charlie) Chan

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Research interests. Robot learning and manipulation. I work on making learned manipulation policies hold up on real robots: under partial observability, in dynamic tasks where reaction time matters, and with sensing beyond vision — Wi-Fi fingerprints for hidden spatial state in my M.S., tactile and force for hidden contact state now. I own the full stack: hardware, data collection, training, and real-robot evaluation.

EDUCATION

M.S. in Artificial Intelligence, National Yang Ming Chiao Tung University (NYCU), Taiwan Sep 2020 – Jun 2022
GPA 4.26/4.30 · Presidential Award (Book Scholarship) · Advisors: Prof. Yu-Chee Tseng, Prof. Jen-Jee Chen
Thesis: zero-shot Wi-Fi fingerprint inpainting with generative adversarial networks · Exchange semester, University of Agder, Norway (Spring 2022), research supervised by Prof. Frank Y. Li

B.S. in Electrical Engineering and Computer Science, National Tsing Hua University (NTHU) Sep 2015 – Jun 2019

RESEARCH EXPERIENCE

Research Assistant, Robot Learning Lab (RLLab), National Taiwan University Mar 2026 – Present
Advisor: Prof. Shao-Hua Sun

- Led the real-world experiments for *Video Replanning via Latent Embedding Refinement and Rejection-Based Sampling* (RELIC, formerly ISE; arXiv:2510.17315, second author): designed and built four physical tasks with hidden physical state (door push/pull mechanism, block friction, bar center of mass, ball mass) on a Piper arm and a WidowX arm with a single webcam, collected the teleoperated training videos (18–25 per task), 3D-printed the fixtures and task objects, and ran the 20-trial-per-task evaluations: RELIC 63.8% vs. 30.0% for the AVDC baseline, improving on every task.
- Independent study of imitation-learning policies for dynamic manipulation on the lab's PiPER arm: evaluation-protocol design, controlled ablations, and system identification for sim-to-real transfer (ongoing, unpublished).
- **Tactile representation learning** for manipulation: whether perception-level gains of tactile encoders transfer to policy-level success, using unpaired onboarding of new tactile sensors as controlled representation variation.

M.S. Research, PAIRLab, NYCU, in collaboration with XunweiTech Sep 2020 – Jun 2022
Advisors: Prof. Yu-Chee Tseng, Prof. Jen-Jee Chen

- Zero-shot Wi-Fi fingerprint inpainting: recovering radio maps for unsurveyed regions (hidden spatial state from non-visual signals). Developed a GAN-based inpainting model with a shared-weight discriminator on (location, fingerprint) pairs, improving positioning accuracy by 20% over benchmarks and 10% in a real-site deployment.
- Extended to Wi-Fi/magnetic fingerprint augmentation with graph-based generative models, improving indoor-positioning robustness and enabling real-world deployment with the industry partner.

Student Researcher, Behavioral Informatics & Interaction-Computation (BIIC) Lab, NTHU Sep 2017 – Jun 2018

- Textual modality of a multimodal (speech, text, image) doctor–child picture-book reading study for autism screening: built a Python NLP pipeline (Jieba segmentation, keyword extraction) and cross-validated classification experiments to identify discriminative linguistic features.

PUBLICATIONS

1. **Y. Chan**, P.-Y. Lin, Y.-Y. Tseng, J.-J. Chen, Y.-C. Tseng. "Learning-Based WiFi Fingerprint Inpainting via Generative Adversarial Networks." *Proc. 33rd IEEE International Conference on Computer Communications and Networks (ICCCN)*, Kailua-Kona, HI, USA, 2024, pp. 1–7. arXiv:2407.18915.
2. P.-C. Ko, Y. Chan, Y.-H. Fu, H.-J. Yeh, C.-R. Chen, D.-C. Tsai, W.-C. Ma, Y. Du, J. Mao, S.-H. Sun. "Video Replanning via Latent Embedding Refinement and Rejection-Based Sampling." *arXiv:2510.17315 (v3, Oct 2026); under review at ICRA 2027.* Previously titled "Implicit State Estimation via Video Replanning." Second author; led the real-world experiments.

OPEN-SOURCE & ENGINEERING

- **RLLxLerobot** (github.com/charlie8612/RLLxLerobot) — the lab's end-to-end imitation-learning stack: teleoperation, dataset recording, training and evaluation scripts, waypoint tooling, and hardware configuration for an AgileX Piper arm with a ROBOTIS OMY leader.
- **LeRobot hardware plugins** — authored and published `lerobot_robot_piper` (AgileX Piper follower; safety-first: no startup lunge, slew-rate limiting), `lerobot_teleoperator_omy` (ROBOTIS OMY-L100 leader via direct Dynamixel Protocol 2.0, no ROS 2), and `lerobot_omy_piper_retarget` (OMY → Piper retargeting processor). Dynamixel motor support merged upstream into Hugging Face LeRobot.

- **Fabrication** — design and 3D-print the physical side of the lab's real-robot experiments: fixtures, mounts and task objects for the RELIC submission; a stiffness tester for the Piper gripper published on MakerWorld.

INDUSTRY EXPERIENCE

AI Application Engineer, Novatek Nov 2024 – Mar 2026

- Designed and optimized a dual-channel real-time sound event localization and detection (SELD) model and its inference pipeline for low-latency deployment on resource-constrained embedded systems; built an on-device streaming speech-to-text translation prototype.
- Surveyed and prototyped VLM-based auto-labeling for object-detection datasets; evaluated parameter-efficient fine-tuning (e.g., LoRA) for adapting large models on edge devices, with deployment recommendations adopted by the team.

Wi-Fi R&D Software Engineer, MediaTek Sep 2022 – Nov 2024

- Owned the Fine Timing Measurement (FTM) ranging feature end-to-end: rebuilt the firmware architecture, improving distance-measurement accuracy from 2–3 m to within 1 m on production Wi-Fi chipsets; RMAC firmware module owner, drove an architectural refactor unifying code across product lines.
- 100% pass rate in ISTA (Wi-Fi Alliance) certification tests; automated the regression and certification workflow, cutting manual testing time by 80%; represented MediaTek at the Wi-Fi Alliance PlugFest (San Francisco) and an IoT interoperability event.

Co-Founder & Developer, Tera Thinker (AI education startup) Jan 2021 – Present

- Built an adaptive AI tutoring platform for high-school students: topic-level question recommendation, a knowledge-tracing module modelling student mastery over time, and automated essay scoring for Taiwanese standardized tests; now reaching about 40% of high schools in Taiwan.

AWARDS

Presidential Award (Book Scholarship), NYCU	2020
Top 25 Finalist, CHT 5G Application Competition	2021
Foxconn–NYCU Joint Research Center Scholarship	2021
MediaTek vAward (×2)	2023–2024

SKILLS

Robotics & ML: imitation learning / diffusion policies, generative models (GAN, VAE), LLM/VLM, speech & audio ML, system identification, real-robot evaluation; PyTorch, TensorFlow, LeRobot, MuJoCo, ROS

Systems: Python, C/C++, Verilog, embedded firmware, CAN/Dynamixel hardware integration, CAD & 3D printing; React

Languages: Mandarin Chinese, Taiwanese (native); English (fluent)

TEACHING & SERVICE

Service-Learning Teaching Assistant, NTHU (2017–2018): weekly English lessons for underprivileged students in Qionglin for 2+ years; mentored volunteers · Volunteer STEAM tutor, community tutoring program (2022)

REFEREES

Prof. Shao-Hua Sun, Robot Learning Lab, National Taiwan University · Prof. Yu-Chee Tseng, National Yang Ming Chiao Tung University · Prof. Frank Y. Li, University of Agder (frank.li@uia.no)